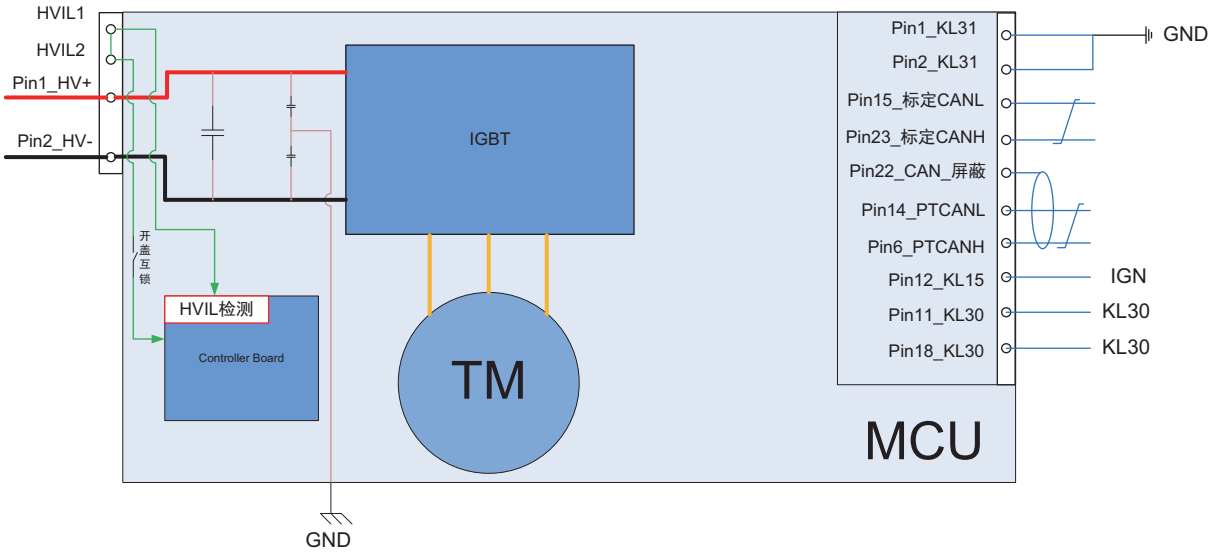
P1909-00、 P190A-1D、 P190B-1D、 P190C-1D

Fault diagnosis code
P1909-00：Three-phase current and overrange
P190A-1D：U-phase current sampling zero drift over range
P190B-1D：V-phase current sampling zero drift over range
P190C-1D：W phase current sampling zero drift over range
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • PEU fault • High voltage insulation fault • The current circuit of PEU fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
P1909-00
The absolute value of three phases current exceeds the upper threshold （70A）
P190A-1D
The absolute value of U-phase current zero drift exceeds the upper threshold （30A）
P190B-1D
The absolute value of V-phase current zero drift exceeds the upper threshold （30A）
P190C-1D
The absolute value of W-phase current zero drift exceeds the upper threshold （30A）
To set the effect of a fault code condition
When vehicle lost power, there is forced power at high speed

Circuit diagram



EV901001

Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of PEU for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the PEU with diagnostic apparatus. 4 Is there any other fault code except for P1909-00、P190A-1D、 P190B-1D、 P190C-1D?
		YES → Refer to:DTC Summary list(PEU).
NO ↓		
Step 3	Check the connecting cable of PEU.	
		1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Disconnect the harness plug battery pack high voltage. 4 Disconnect the high voltage harness plug of PEU. 5 Check whether the PEU connection cable is rotten, damaged, and water enters. 6 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 4	Check the insulation of PEU.	
		1 Use a megohm meter to check the insulation resistance of the PEU housing and grounding according to the procedure in the maintenance manual. **Standard value: >20MΩ** 2 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		

Step 5	Check the insulation of driving motor.
	1 Use a megohm meter to check the insulation resistance of the driving motor housing and grounding according to the procedure in the maintenance manual. **Standard value: >20MΩ** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	
Step 6	Perform initialization.
	1 Turn the ignition switch to OFF. 2 Connect the fault diagnosis instrument to the diagnosis interface. 3 Turn the ignition switch to ON. 4 The governor was initialized with the diagnostic instrument. 5 Can the governor be initialized and is the governor restored?
	NO → Repair or change the governor.
YES ↓	
Step 7	Check PEU internal circuit.
	1 Turn the ignition switch to OFF. 2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual. 3 Open the PEU end cover. 4 Check the PEU internal circuit and components for obvious damage or . ablation. 5 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 8	Test whether PEU is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB1002 of PEU. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. 4 Fly out of pins-install new pins as needed. 5 Reconnect the PEU plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 6 Run the system and determine if the problem persists?
	YES → Change the PEU Module
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems.	
Next Step ↓	
Diagnosis end.	